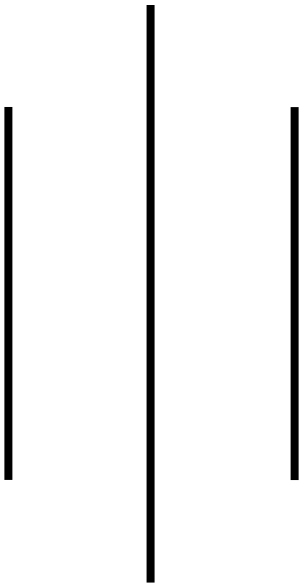


Tribhuvan University

Faculty of Humanities and Social Sciences

Danfe college

Sinamangal, Kathmandu



Department of Bachelor of Computer Applications (BCA)

LAB REPORT OF CACS 402: Cloud Computing

Submitted by:

Name: Cherub Singh Thing Tamang

Level: 7th semester

Submitted to:

Name: Jiwan Khadka

Signature:

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Sinamangal, Kathmandu



Lab Report

On

Cloud Computing

(CACS 402)

For the partial fulfillment of Bachelor's Degree of Computer Applications

SUBMITTED BY:

Cherub Singh Thing Tamang

BCA, seventh Semester

Symbol No.: 92002108

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SUBMITTED TO:

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Department of Bachelor in Computer Application

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Sinamangal, Kathmandu

This is to certify that the Lab Reports prepared entitled **"CLOUD COMPUTING"** is an academic work done by **"Cherub Singh Thing Tamang"** submitted in the partial fulfillment of the requirements for the degree of **Bachelor of Computer Application** at Faculty of Humanities and Social Science, Tribhuvan University under my guidance and supervision. To the best of my knowledge, the work performed by him/her in the lab reports is his/her own creation.

Signature
Subject Teacher
BCA

Signature
HOD/Coordinator/Principal
BCA

Signature
Internal Examiner

Signature
External Examiner

ACKNOWLEDGEMENT

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Finally, I would like to acknowledge the support of my family and friends, who provided encouragement and motivation throughout this project.
Thank you all for your contributions to the success of this lab report.

Name: - Cherub Singh Thing Tamang
Registration No:- 6-2-920-7-2

Lab No.	Topic Name	Page No.	Remarks
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2	Enlist various companies in cloud business and the corresponding services provided by them and tag them under SaaS, PaaS and IaaS.	5 to 8	
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4	Install a C compiler in the virtual machine created using virtual box/VMware and execute Simple Programs using Terminal.	12 to 13	
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Class: BCA(VII) **Subject: Cloud Computing**

Experiment No. 1

Aim: To study in detail about cloud computing.

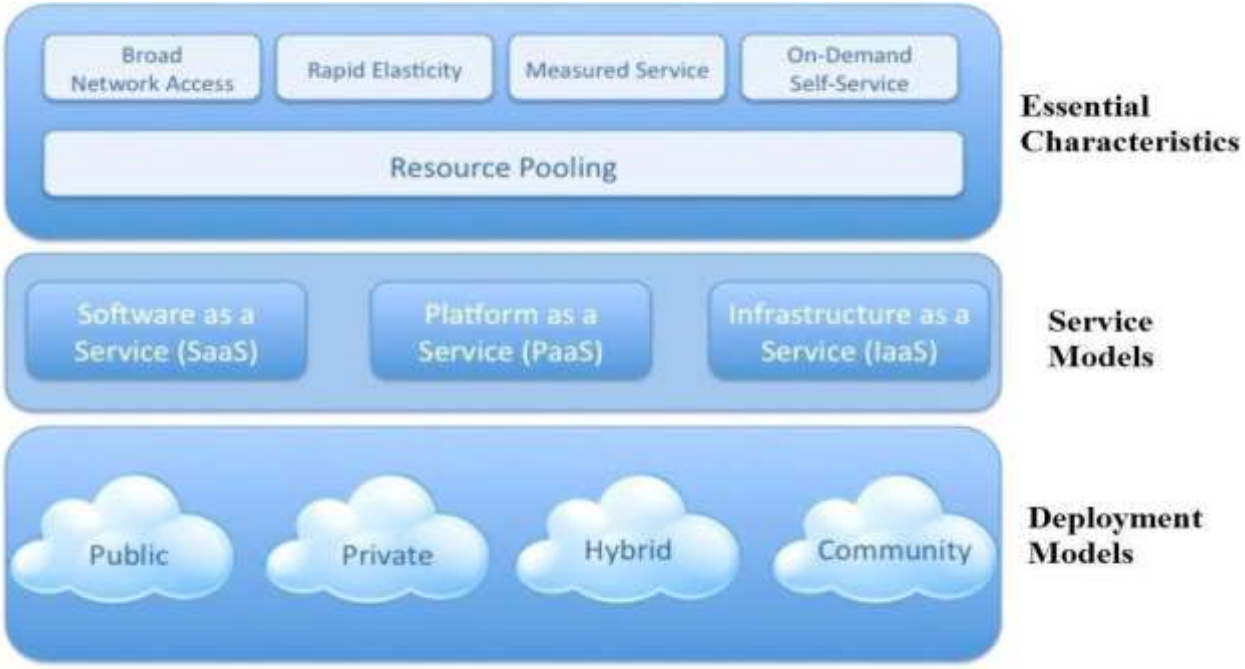
Theory:

The term *cloud* has been used historically as a metaphor for the Internet. This usage was originally derived from its common depiction in network diagrams as an outline of a cloud, used to represent the transport of data across carrier backbones (which owned the cloud) to an endpoint location on the other side of the cloud. This concept dates back as early as 1961, when Professor John McCarthy suggested that computer time-sharing technology might lead to a future where computing power and even specific applications might be sold through a utility-type business model. 1 This idea became very popular in the late 1960s, but by the mid-1970s the idea faded away when it became clear that the IT-related technologies of the day were unable to sustain such a futuristic computing model. However, since the turn of the millennium, the concept has been revitalized. It was during this time of revitalization that the term *cloud computing* began to emerge in technology circles. Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction .A Cloud is a type of parallel and distributed system consisting of a collection of inter-connected and virtualized computers that are dynamically provisioned and presented as one or more unified computing resource(s) based on service-level agreements established through negotiation between the service provider and consumers.

When you store your photos online instead of on your home computer, or use webmail or a social networking site, you are using a cloud computing service. If you are in an organization, and you want to use, for example, an online invoicing service instead of updating the in-house one you have been using for many years, that online invoicing service is a ―cloud computing service. Cloud computing is the delivery of computing services over the Internet. Cloud services, allow individuals and businesses to use software and hardware that are managed by third parties at remote locations. Examples of cloud services include online file storage, social networking sites, webmail, and online business applications. The cloud computing model allows access to information and computer resources from anywhere. Cloud computing provides a shared pool of resources, including data storage space, networks, Computer processing power, and specialized corporate and user applications.

Architecture

- Cloud Services Models.
- Cloud Deployment Models.
- Essential characteristics of cloud computing.



NIST Visual Model of Cloud Computing Definition

Cloud Service Models

- Cloud **S**oftware as a **S**ervice (SaaS)
- Cloud **P**latform as a **S**ervice (PaaS)
- Cloud **I**nfrastructure as a **S**ervice (IaaS)

Infrastructure as a Service (IaaS) :--

 The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources.

 Consumer is able to deploy and run arbitrary software, which can include operating systems and applications.

 The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems; storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).

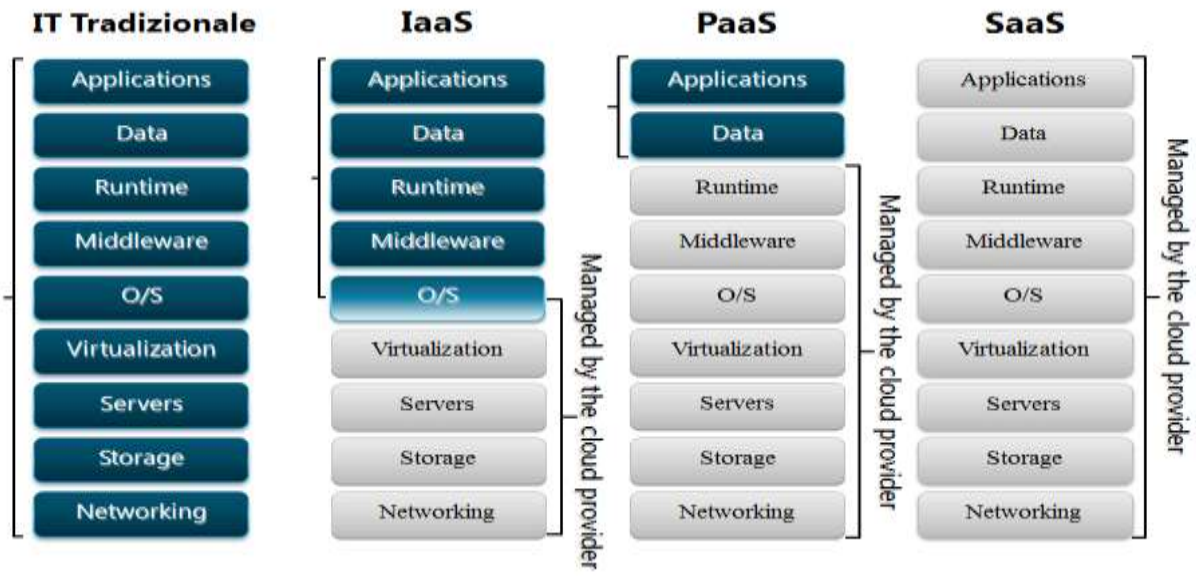
Platform as a Service (PaaS):--

The capability provided to the consumer is to deploy onto the cloud infrastructure consumer created or acquired applications created using programming languages and tools supported by the provider.

The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations

Software as a Service (SaaS):

- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure.
- The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email)
- The consumer does not mange or control the underlying cloud infrastructure including network, servers, operating system, storage, or even individual application capabilities, with the possible exception of limited user specific application configuration settings.



Cloud Deployment Models:

- Public
- Private
- Community
- Hybrid

❖ **Public Cloud:** The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.

- **Private Cloud:** The cloud infrastructure is operated solely for a single organization. It may be managed by the organization or a third party, and may exist on-premises or off-premises.
- **Community Cloud**: The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, or compliance considerations). It may be managed by the organizations or a third party and may exist on-premises or off-premises.
- **Hybrid Cloud**: The cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).

ESSENTIAL CHARACTERISTICS:--

 On-demand self-service:--A consumer can unilaterally provision computing capabilities such as server time and network storage as needed automatically, without requiring human interaction with a service provider.

 Broad network access:--Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs) as well as other traditional or cloud based software services.

 Resource pooling:--The provider‘s computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources
dynamically assigned and reassigned according to consumer demand.

 Rapid elasticity:--Capabilities can be rapidly and elastically provisioned in some cases automatically - to quickly scale out; and rapidly released to quickly scale in. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.

 Measured service:--Cloud systems automatically control and optimize resource usage by leveraging a metering capability at some level of abstraction appropriate to the type of service. Resource usage can be monitored, controlled, and reported - providing transparency for both the provider and consumer of the service.

Conclusion:

Thus, we have studied in detail about overview of cloud computing

Class: BCA(VII) **Subject: Cloud Computing**

Experiment No. 2

Aim: To enlist various companies in cloud business and the corresponding services provided by them and tag them under SaaS, PaaS and IaaS.

Theory:

The cloud refers to how and where data is stored – perhaps more importantly, where it isn't. The cloud allows software and services to run on the internet instead of only locally on one device because the data is stored remotely across various servers.

IaaS, PaaS, and SaaS are three main categories of cloud computing. Cloud computing is using a network of different servers that host, store, manage, and process data online – "in the cloud".

Definition of each service are mention below:

- **IaaS** (Infrastructure as a Service): IaaS products allow organization to manage their business resources – such as their network, servers, and data storage – on the cloud.
- **PaaS** (Platform as a Service): PaaS products allow businesses and developers to host, build, and deploy consumer-facing apps.
- **SaaS** (Software as a Service): By far the most common cloud service, SaaS products offer both consumers and businesses cloud-based tools and applications for everyday use.

We can access all three services via Internet browser or online apps available on different devices. A great example is Google Docs, which allows the team to collaborate online instead of needing to work on one Microsoft Word document and send it around to each other.



Compare these to on-premise software, which is installed locally on a server or device at an organization's physical location.

IaaS (Infrastructure as a Service)

IaaS, or Infrastructure as a Service, is a cloud-based service that allows resources to be delivered to organizations virtually (or through the cloud). IaaS tools help organizations build and manage servers, networks, operating systems, and data storage without needing to buy hardware.

IaaS customer can control their data infrastructure without physically managing it on-site. Instead, they store data on the servers of IaaS providers, and use a dashboard or API (Application Programming Interface) to access and manage their resources.

IaaS Examples

- **Amazon Web Services (AWS)**: AWS is overseen by Amazon and is used for on demand cloud computing and purchase in a recurring subscription basis. AWS helps companies to store data and deliver content over the Internet.
- **Microsoft Azure**: Microsoft Azure is cloud computing IaaS product that allows for building, testing, and managing applications through a network of Microsoft Data Centers.
- **Google Cloud:** Google Cloud is an IaaS platform that businesses can use to natively run Windows, Oracle, SAP. Additionally, a business can manage its enterprise database and use AI solutions to increase operational efficiency within the firm.
- **IBM Cloud:** IBM Cloud is another IaaS product that allows businesses to "allocate your computer, network, storage and security resources on demand." In other words, businesses only use resources when needed, increasing efficiency.

Platform as a Service (PaaS)

A PaaS or Platform as a Service, provides developers with a framework they can use to build custom applications. PaaS doesn't deliver software over the Internet, but rather a platform that developers can use to create online software and apps.

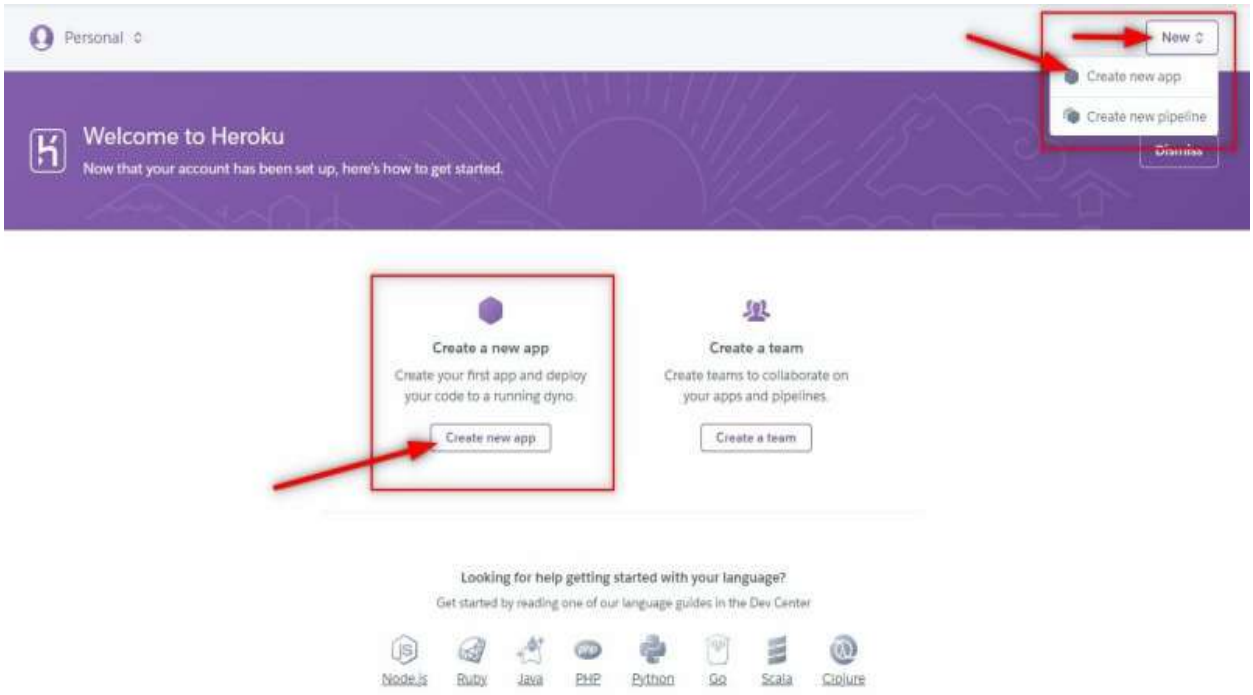
What does PaaS do?

PaaS products let developers build custom applications online without having to deal with data serving, storage, and management.

We can think of PaaS as a scaled-down version of IaaS. It still provides customers with servers and data centers to store their information (in this instance, an app), but its customer is a developer creating an app that will then be delivered over the Internet to consumers. SaaS applications are developed on PaaS platforms.

PaaS Delivery

PaaS provides a secure platform for developers where they can create software and apps for consumer use. Developers access PaaS tools over the web. For instance, the Heroku dashboard, prompted to create an app.



PaaS Examples:

- **Google App Engine:** Google App Engine allows developers to build and host web applications in cloud-based data centers that Google manages.
- **Kinsta:** Kinsta provides an Application, Database and Managed WordPress Hosting solutions that make it quick and easy to deploy any web applications in minutes, without worrying about the hosting infrastructure.

- **Red Hat OpenShift:** Red Hat OpenShift is an on-premises containerization PaaS software.
- **Heroku:** Developers can use this PaaS tool to build, manage, and grow consumer facing-apps.
- **Apprenda:** Apprenda is a PaaS product that allows developers and businesses to host an entire application portfolio. Build and deploy applications of all types on this platform.
- **GitHub, GitLab, Kubernetes, Docker etc.**

Software as a Service (SaaS)

SaaS, or Software as a Service, refers to cloud-based software that is hosted online by a company, is available for purchase on a subscription basis, and is delivered to buyers via the Internet.

SaaS Examples:

- **HubSpot:** HubSpot is a CRM, marketing, sales, and service SaaS platform that businesses use to connect with and retain customers.
- **JIRA:** JIRA is a project management software that's delivered by Atlassian and can be purchase on subscription basis by customers.
- **DropBox:** DropBox is a file-sharing SaaS tool that allows multiple users within a group or organization to upload and download different files.
- **DocuSign:** DocuSign is a SaaS product that businesses use to send contracts and other documents that require signatures.

Other Examples:



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Experiment No. 3

Aim: To Create a Virtual Machine for Ubuntu/Linux or Windows on VMware Workstation or Oracle's Virtual Box.

Theory:

VMware is a software platform that allows you to create and run virtual machines (VMs). A virtual machine is like a computer inside a computer, where you can run different operating systems (e.g., Linux, Windows) without affecting your main system.

VMware helps developers, IT professionals, and businesses use one physical machine to operate multiple "virtual" computers, saving hardware costs and improving resource efficiency.

Here's a step-by-step guide to creating a Virtual Machine (VM) for Linux on VMware Workstation:

Step 1: Install VMware Workstation

1. **Download and Install VMware Workstation**:
    - Visit the VMware official website and download VMware Workstation Pro or VMware Workstation Player.
    - Install it on your system by following the setup instructions.

Step 2: Create the Virtual Machine

1. **Launch VMware Workstation**:

   Open VMware Workstation from your system.
2. **Create a New Virtual Machine**:
    - Click **File > New Virtual Machine** or **Create a New Virtual Machine** from the home screen.
3. **Choose the Configuration**:
    - Select **Typical (recommended)** and click **Next**.
4. **Select the Installation Media**:
    - Choose **Installer disc image file (ISO)**.
    - In the same step, use the **Download ISO** option provided by VMware Workstation to download the latest Ubuntu ISO image directly.
    - Wait for the download to complete, and then VMware will automatically detect the ISO file.

5. **Guest Operating System**:
    - Select **Linux** as the guest OS type.
    - Choose **Ubuntu 64-bit** as the version.
    - Click **Next**.
6. **Name the Virtual Machine**:
    - Enter a name for your VM (e.g., "Ubuntu VM").
    - Specify the location where the VM files will be stored.
7. **Specify Disk Capacity**:
    - Set the disk size (e.g., 20 GB or more).
    - Select **Store virtual disk as a single file** (recommended for performance).
    - Click **Next**.
8. **Customize Hardware (Optional)**:
    - Click **Customize Hardware** to adjust settings like:
        - **Memory (RAM)**: Allocate at least 4 GB for better performance.
        - **Processors**: Assign 2 or more cores if your system allows.
        - **Network Adapter**: Use **NAT** for internet access.
    - Click **Close** after customization.
9. **Finish the Setup**:
    - Review your settings and click **Finish**.

Step 3: Power On the Virtual Machine

1. **Start the VM**:
    - Select your newly created VM and click **Power on this virtual machine**.
2. **Begin Ubuntu Installation**:
    - Follow the on-screen instructions to install Ubuntu:
        - Select your **language**.
        - Connect to the network (if needed).
        - Choose **Normal Installation** or **Minimal Installation** based on your need.
        - Partition the disk (default settings are usually fine).
        - Create a **username** and **password**.

3. **Complete Installation**:
    - After the installation is complete, restart the VM when prompted.

After successful installation here is the screenshots:



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Experiment No. 4

Aim: To install C compiler and running a simple C program in the virtual machine using VMware.

To install C compiler and running a simple **hello.c** program.

Step: 1

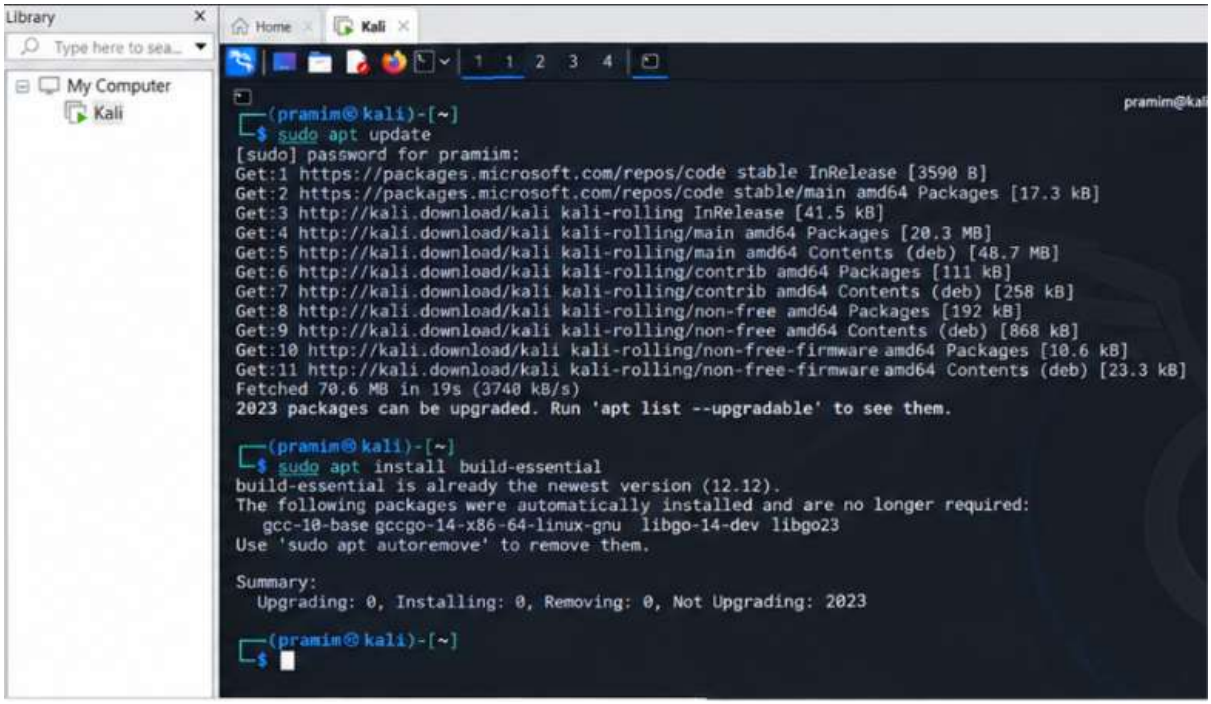
- First We have to open the terminal

Step: 2

- Then type the command: **sudo apt update** => This update our sudo related library in the virtual machine.

Step: 3

- Then type Command: **sudo apt install build-essential**
    - After running this program, we get this result



Step: 4

- Now, create a file name hello.c by executing the command: **touch hello.c** in the terminal in the destop by changing the directory like: running the command **cd desktop**

Step: 5

- Now, run the command: **gedit hello.c.** Then, the editor will be open and writing a simple C program in it.

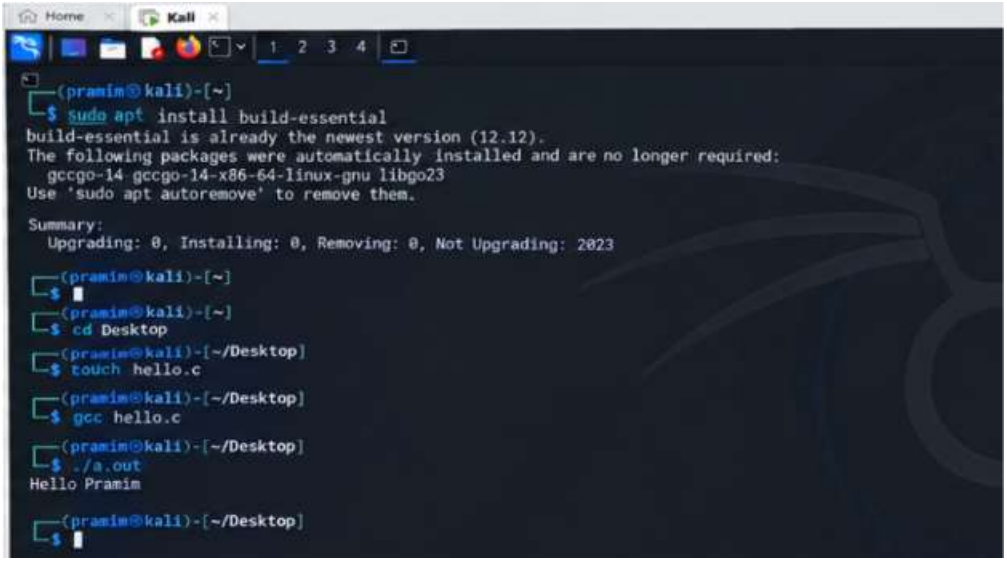
Step: 6

- After that exit from there, and write **gcc hello.c –o hello** command**.** This command help to compile our c program and create a new folder name **hello**.

Step: 7

- Then run the program by executing the command: **./ hello.**

Here is the result:



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Experiment No. 5

Aim: To perform a multithreading program using C# or Java on Ubuntu(Linux) in Virtual Machine using Terminal or any IDE.

Step: 1

- First we have to open the terminal.

Step: 2

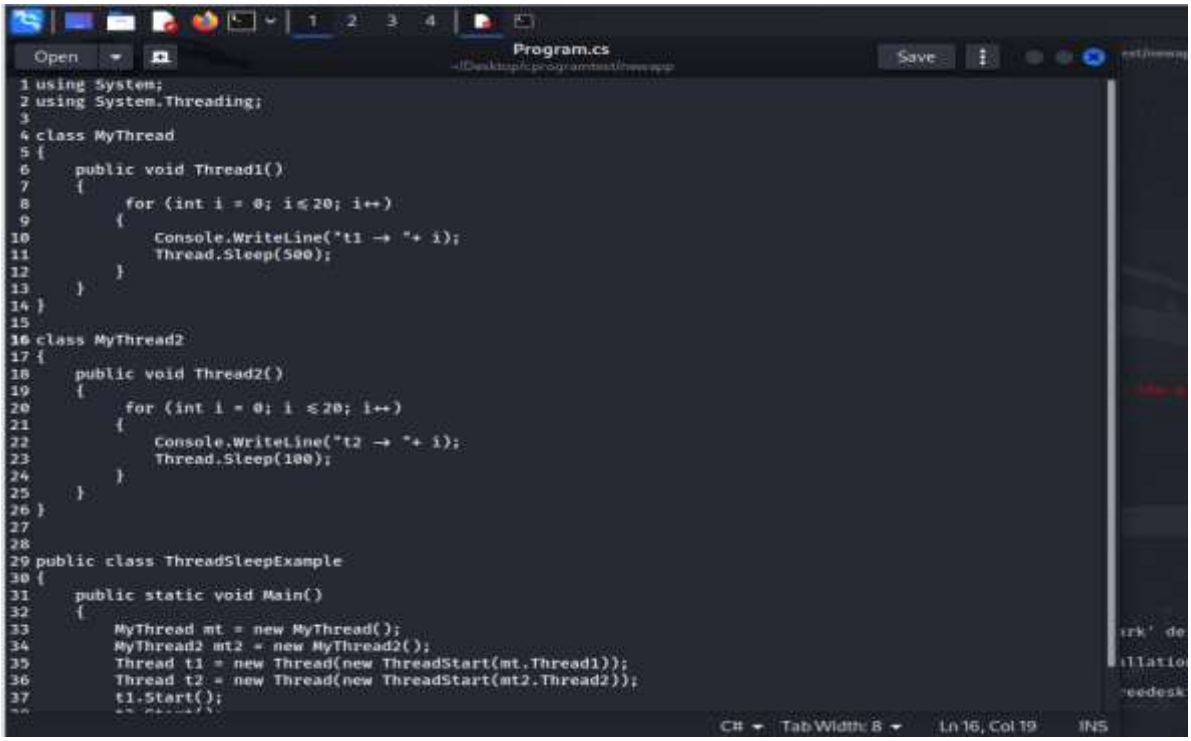
- Now, creating a dotnet folder consisting some .cs files and it's library using such command: **dotnet new console --name newapp**

Step: 3

- Then we get a folder name newapp. So, we locate the directory using command: **cd newapp**

Step: 4

- After change the directory used command: **gedit Program.cs** and add multithreading program in the editor view as given in the image.



Step: 5

- After saving the code then run the program. Here we used the command to run C# code: **dotnet run**

Here, is the result:



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Experiment No. 6

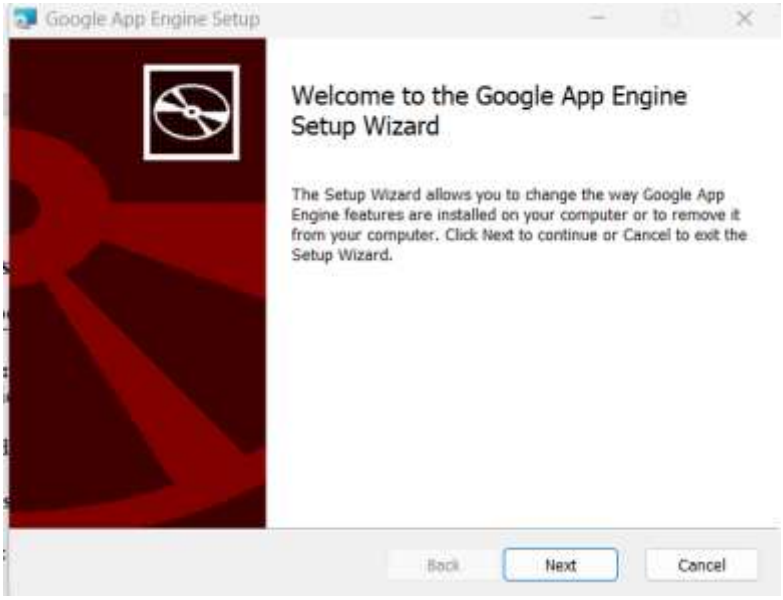
Aim: To Installing Google App Engine and creating a hello world app and other simple web applications using Python/PHP and using GAE launcher to launch the web applications.

For downloading a Google App Engine:

Steps for running a simple web application using GAE launcher.

Step: 1

- Go to the chrome or any browser and download the Google App Engine and install it in your computer.

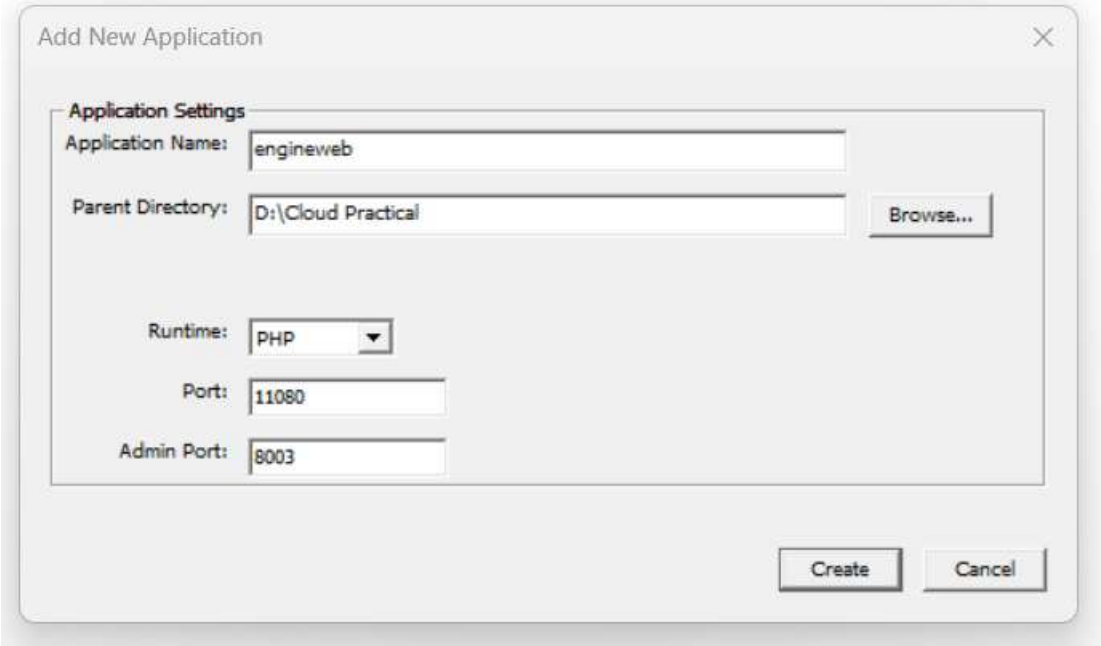


Step: 2

- Then, open the engine launcher and click on file in a menu bar and select to Create New Application or you can use short cut key like CTRL + N for creating new application file.

Step: 3

- Now, add new application details such as application name, select directory that where you want to save, runtime environment, Port which it is provided or you can add it's by your own. And then hit on Create button.



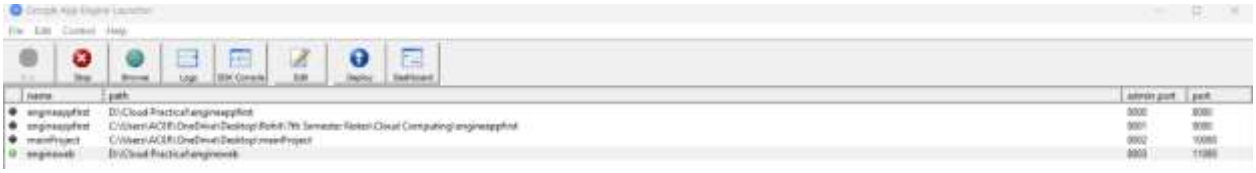
Step: 4

- After we create application, then we get the interface such as:



Step: 5

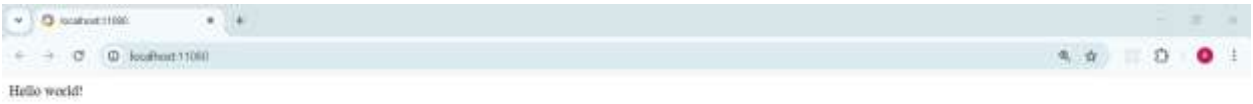
- Then, click on Run button by selecting a specific application. After that server run a specific port:



Step: 6

- Now click on Browse button then It redirects to the default browser and showing our default code which is in php.

Here is the result:



And here is the code:

Yaml source file code:

```
application: engineweb
version: 1
runtime: php55
api_version: 1
threadsafe: yes
handlers:
- url: /favicon\.ico
  static_files: favicon.ico
  upload: favicon\.ico
- url: .*
  script: main.php
```

Php code:

```
<?php
   echo 'Hello world!';
```

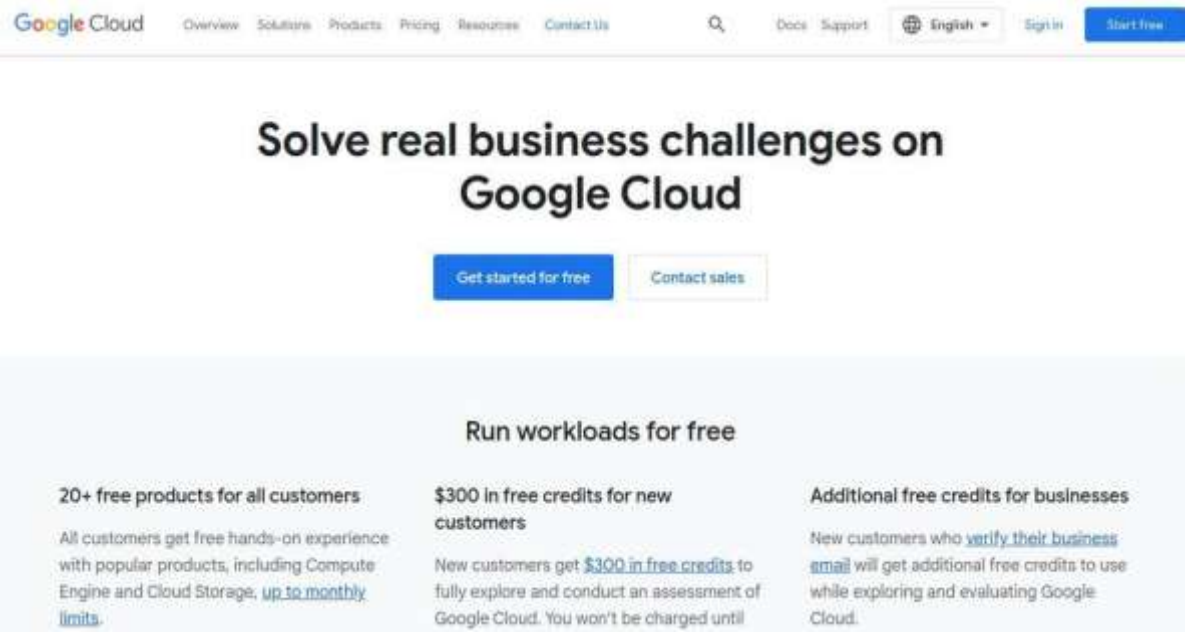
Class: BCA(VII) **Subject: Cloud Computing**

Experiment No. 7

Aim: To Creating a free tier Account for Google Cloud Platform and showing its dashboard.

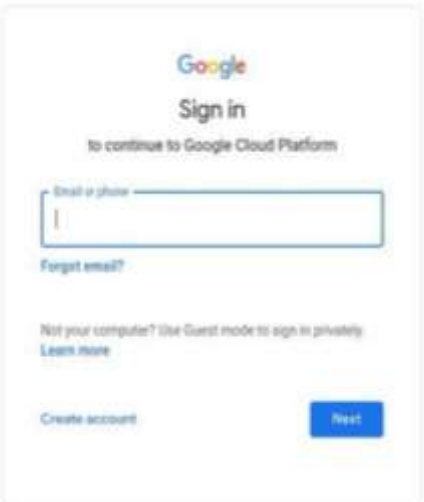
Steps to Create a Free Tier Account on GCP

Step 1: Navigate to the GCP Account creation home page.

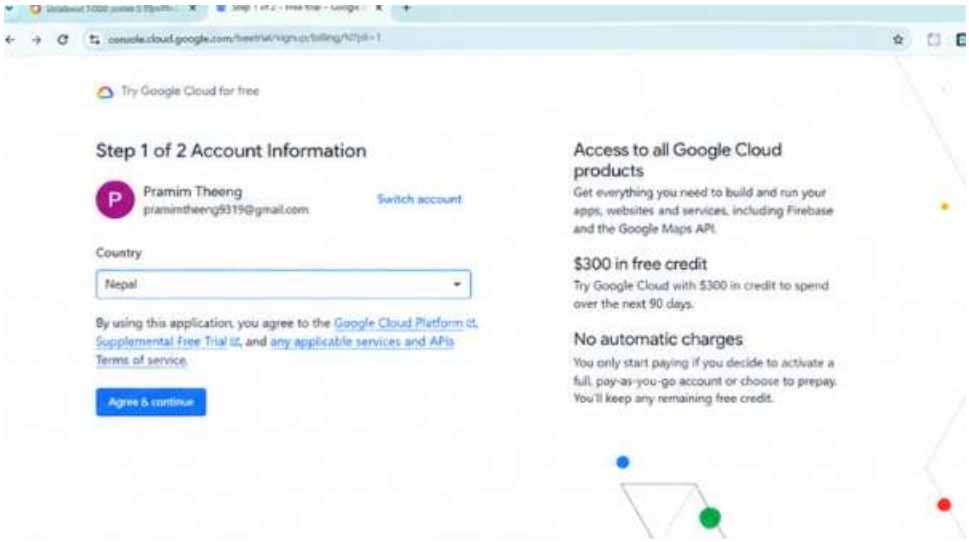


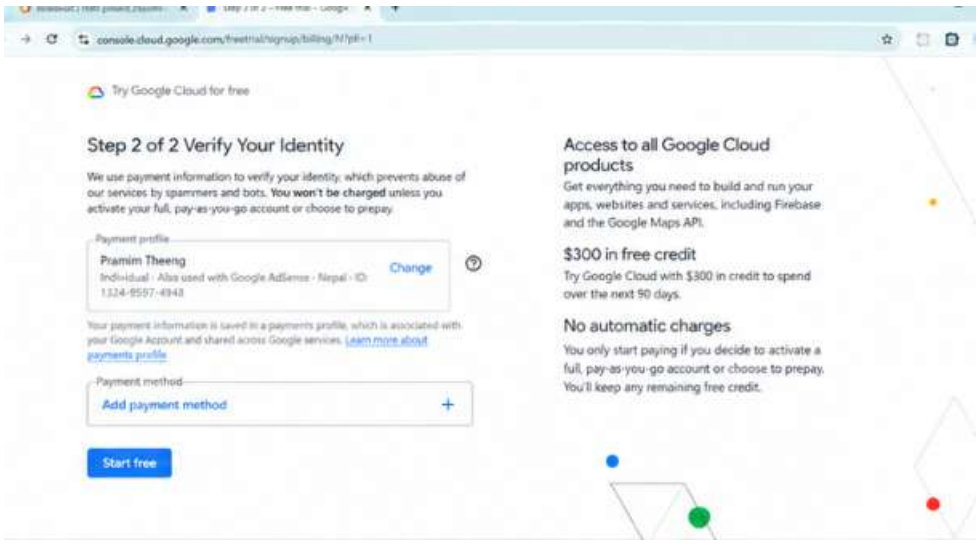
Step 2: Click on the Start free button in the top right corner.

Step 3: Sign in with your Gmail id.



Step 4: After signing in you will be redirected to the GCP Free-trial page. select your country from the drop-down, accept the terms of service, and then click on continue.



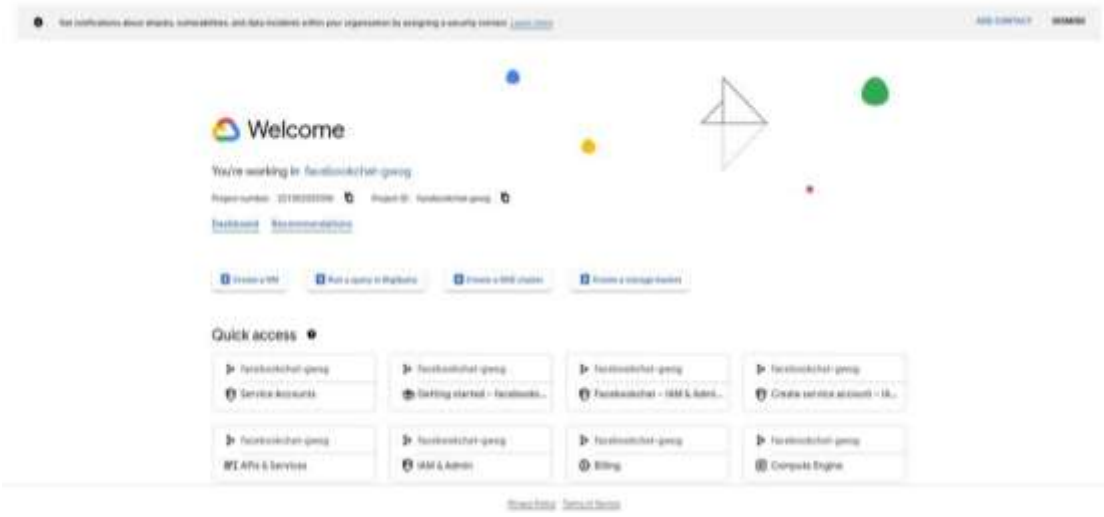


Step 5: Enter the Business name and Enter payment method.

Step 6: Start my Free Trial.

Step 7: Complete the payment and your free trial will start (your money will be refunded in 3 working days).

Step 8: Then after completion of payment you will be redirected to the google cloud console


Welcome
Quick access

Class: BCA(VII) **Subject: Cloud Computing**

Experiment No. 8

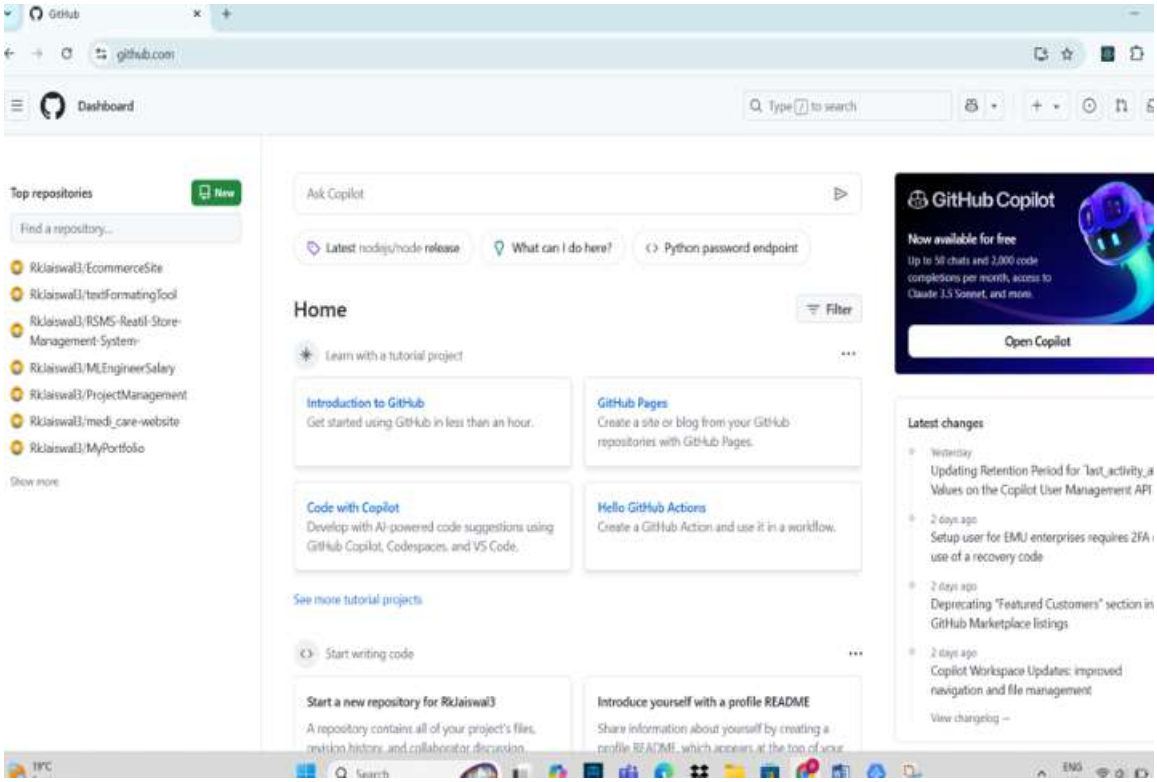
Aim: Create an Account on GitHub.

Signing up for a new personal account

1. Navigate to https://github.com/.
2. Click Sign up.
3. Follow the prompts to create your personal account.

During sign up, you'll be asked to verify your email address. Without a verified email address, you won't be able to complete some basic GitHub tasks, such as creating a repository.

If you're having problems verifying your email address, there are some troubleshooting steps you can take. For more information, see "Verifying your email address."



Class: BCA(VII) **Subject: Cloud Computing**

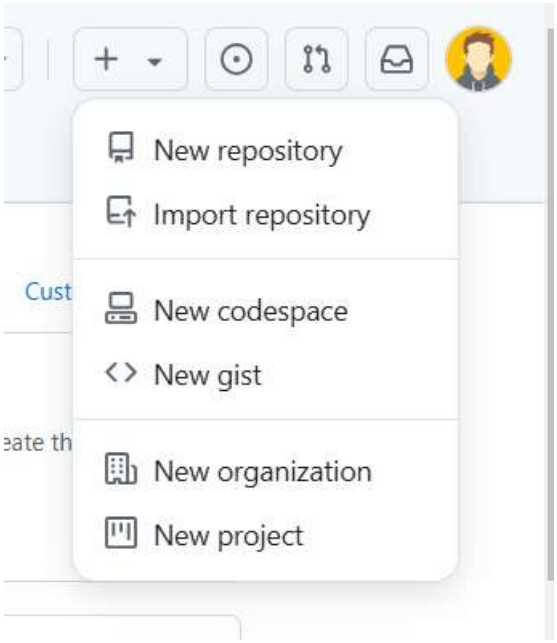
Experiment No. 9

Aim: Host a static Web site on GitHub.

Creating a Website name coolweb.github.io

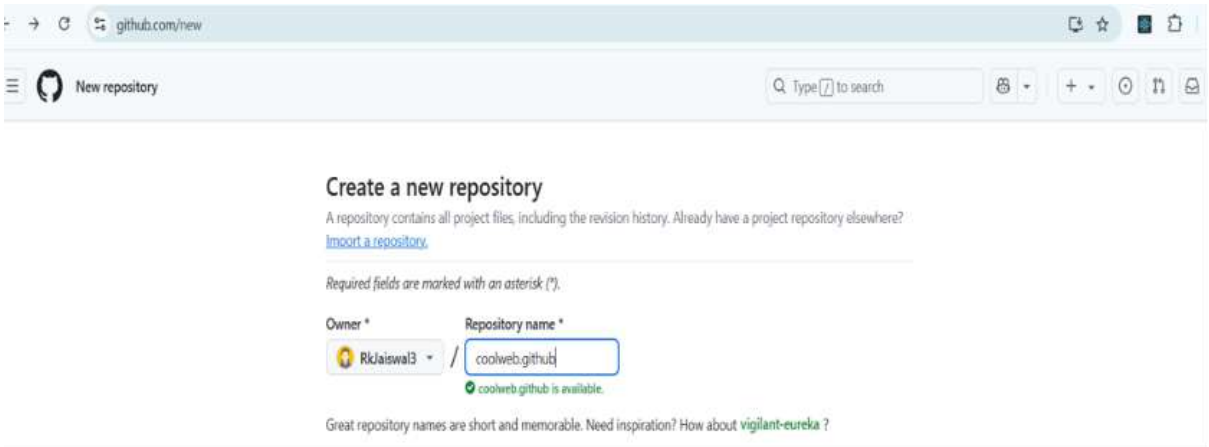
Step: 1

First visit your own GitHub portal and create a new repository by selecting on **New repository**.



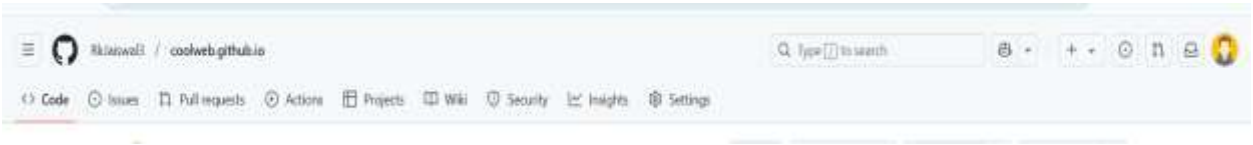
Step: 2

Enter username.github.io as the repository name. Replace username with your GitHub username. For example, if your username is coolweb, the repository name should be coolweb.github.io. And select **Create repository**.

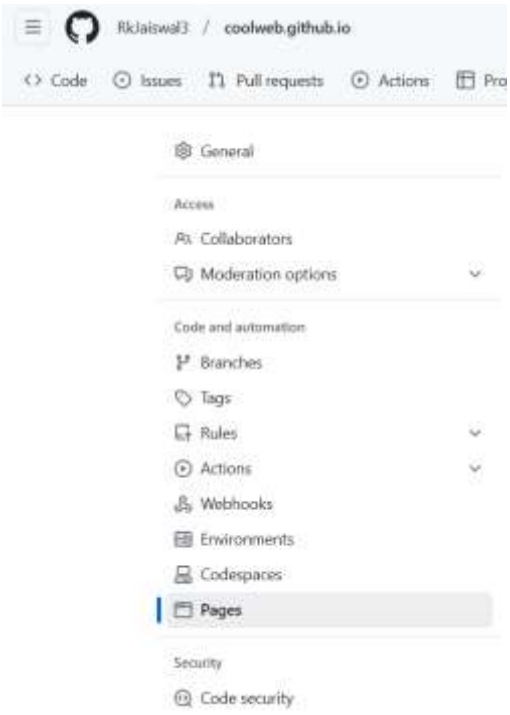


Step: 3

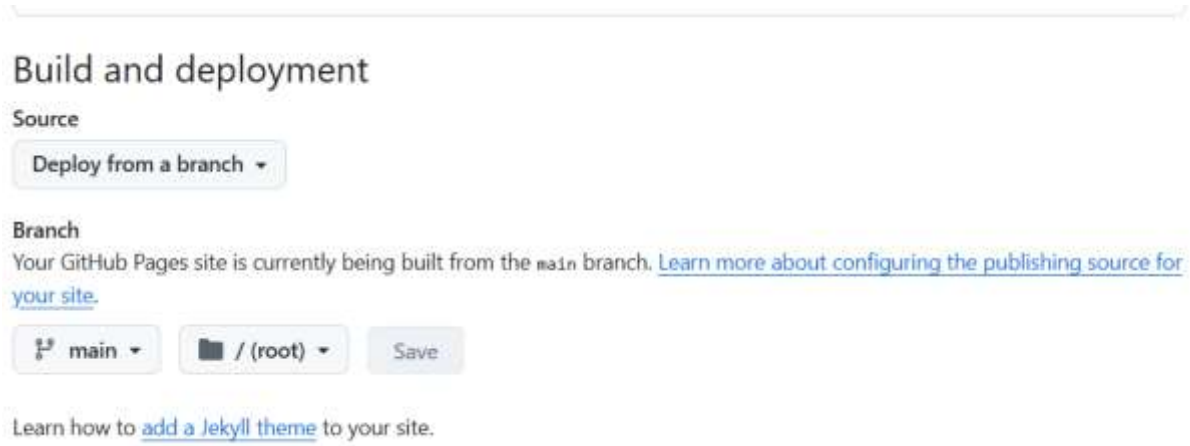
After that you have seen your repository. Under your repository name, click Settings. If you cannot see the "Settings" tab, select the dropdown menu, then click **Settings**.



In the "Code and automation" section of the sidebar, click **Pages**.

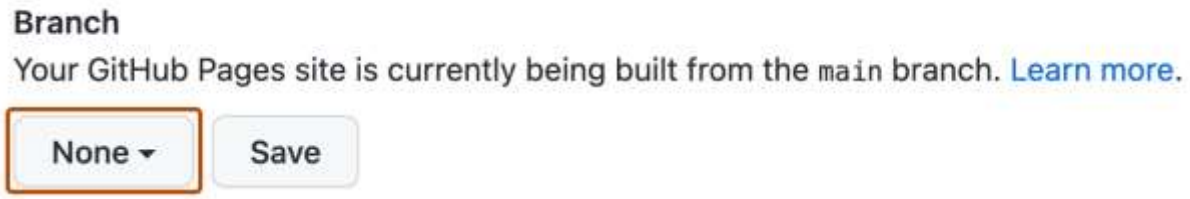


Under "Build and deployment", under "Source", select **Deploy from a branch**.



Step: 4

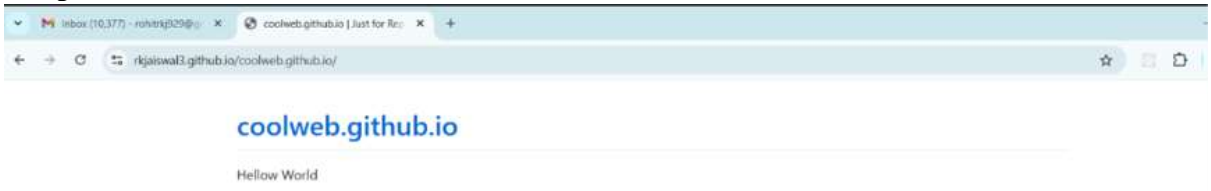
Under "Build and deployment", under "Branch", use the branch dropdown menu and select a publishing source.



Optionally, open the README.md file of your repository. The README.md file is where you will write the content for your site. You can edit the file or keep the default content for now.

Visit username.github.io to view your new website. Note that it can take up to 10 minutes for changes to your site to publish after you push the changes to GitHub.

Output :



Class: BCA(VII) Subject: Cloud Computing

Experiment No. 10

Aim: Case Study of PAAS (Google App Engine) and IAAS (Amazon Web Services)

Introduction

Cloud computing is categorized into different service models, such as Platform as a Service (PaaS) and Infrastructure as a Service (IaaS). This case study compares Google App Engine as a PaaS and Amazon Web Services (AWS) as an IaaS.

Google App Engine (PaaS)

Overview: Google App Engine (GAE) is a Platform as a Service that allows developers to build, deploy, and scale applications without worrying about managing the underlying infrastructure.

Features:

1. **Ease of Use:** Provides a fully managed environment for deploying web applications.
2. **Scalability:** Automatically scales applications based on traffic.
3. **Programming Support:** Supports multiple languages like Python, Java, Node.js, and Go.
4. **Integrated Services:** Comes with built-in services like databases (Datastore), queues (Task Queues), and caching (Memcache).

Advantages:

- No need to manage servers.
- High availability and auto-scaling.
- Pay-as-you-go pricing.

Limitations:

- Limited control over the underlying infrastructure.
- Platform-specific constraints, making it harder to migrate applications.

Use Case: A start up building a web application with limited resources can use Google App Engine to quickly develop and scale without worrying about server management.

Amazon Web Services (IaaS)

Overview: Amazon Web Services (AWS) is an Infrastructure as a Service that provides virtualized computing resources over the internet, enabling users to manage and configure the infrastructure according to their needs.

Features:

1. **Flexible Infrastructure:** Offers virtual machines (EC2), storage (S3), and networking (VPC).
2. **Customization:** Full control over the OS, software stack, and configurations.
3. **Scalability:** Allows manual or automated scaling of resources.
4. **Wide Range of Services:** Provides services like databases (RDS), AI/ML tools, and analytics.

Advantages:

- Complete control over infrastructure.
- Wide range of customizable options.
- Suitable for enterprise-grade applications.

Limitations:

- Requires technical expertise to manage.
- Overhead of configuring and maintaining infrastructure.

Use Case: An enterprise running a large e-commerce website can use AWS to customize the infrastructure for high traffic and integrate advanced analytics tools.

Conclusion

Google App Engine is ideal for developers seeking a quick, scalable solution without managing infrastructure, while AWS is suited for businesses needing customizable and robust infrastructure. Choosing between the two depends on the project's requirements and the technical expertise available.